



MASTG

MASTG-TEST-0251: Runtime Use of Content Provider Access APIs in WebViews

MASTG-TEST-0252: References to Local File Access in WebViews

MASTG-TEST-0253: Runtime Use of Local File Access APIs in WebViews

MASTG-TEST-0258: References to Keyboard Caching Attributes in UI Elements

MASTG-TEST-0289: Runtime Verification of Sensitive Content Exposure in Screenshots During App Backgrounding

MASTG-TEST-0291: References to Screen Capturing Prevention APIs

L2

ANDROID

MASWE-0053

TEST

MASTG-TEST-0258: References to Keyboard Caching Attributes in UI Elements

Overview

This test verifies that the app appropriately configures text input fields to prevent the keyboard from caching sensitive information, such as passwords or personal data.

Android apps can configure the behavior of text input fields using XML attributes in the layout files or programmatically in the code. If the app doesn't use non-caching input types for sensitive data, the keyboard may cache sensitive information.

Steps

- Reverse engineer the app ([🔗 Decompile Java Code](#)).
- Search for XML attributes in the layout files within the `res/layout` directory.
- Search for calls to the `setInputType` method and the input type values passed to it ([🔗 Static Analysis on Android](#)).
- In case the app uses Jetpack Compose, search for calls to `KeyboardOptions` constructors [↗](#) and their parameters in the reversed code ([🔗 Static Analysis on Android](#)). Especially `keyboardType` and `autoCorrect`.

Observation

The output should include:

- All `android:inputType` XML attributes, if using XML for the UI.
- All calls to the `setInputType` method and the input type values passed to it.

Evaluation

The test case fails if there are any fields handling sensitive data for which the app does not use non-caching input types (see section "Non-Caching Input Types" in [🔗 Keyboard Cache](#)).

Best Practices

MASTG-BEST-0019: Use Non-Caching Input Types for Sensitive Fields

Demos

MASTG-DEMO-0064: Uses of Caching UI Elements with semgrep